Cloud Computing Course

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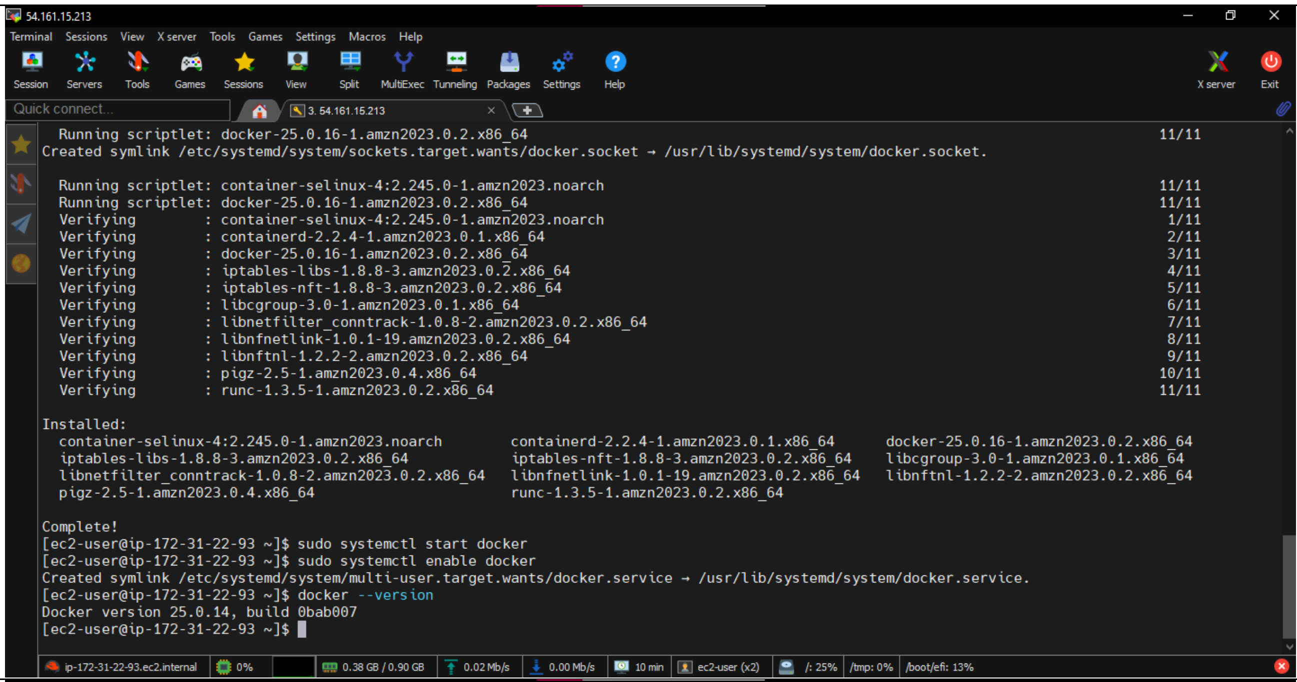
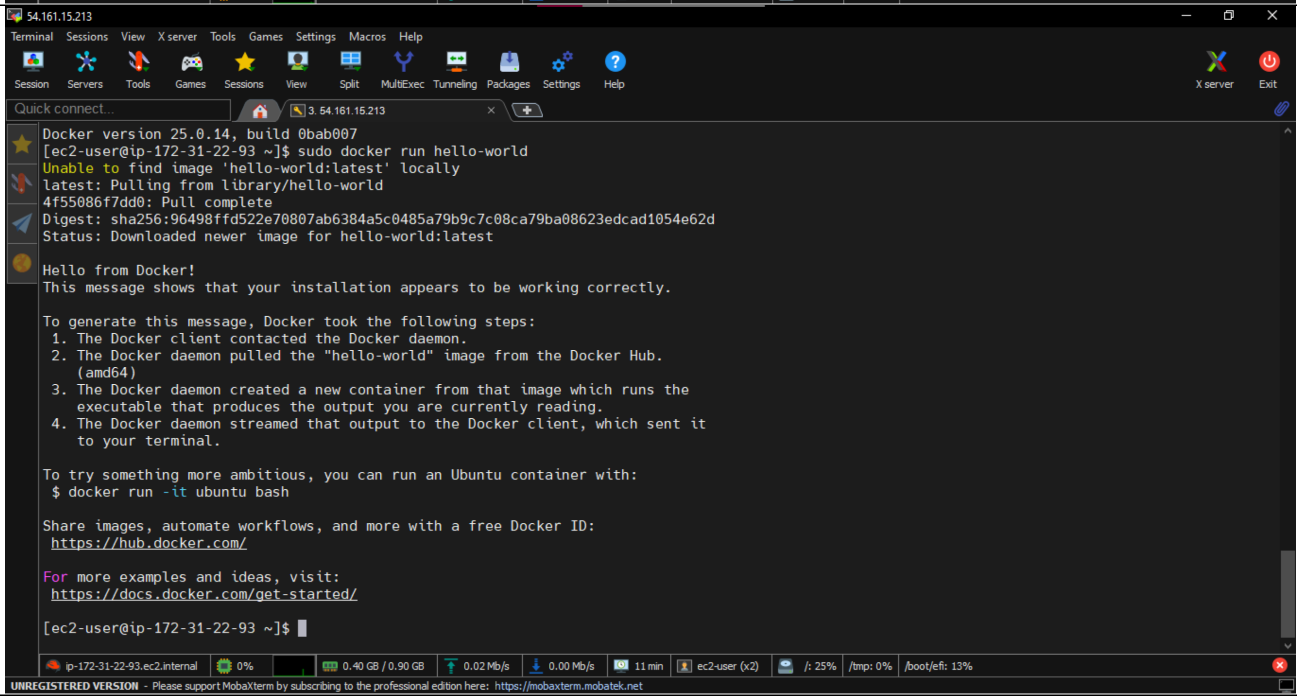
1.1



1.2



1.3

54.161.15.213

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit

Quick connect... 3. 54.161.15.213

```
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

[ec2-user@ip-172-31-22-93 ~]$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c03230f136: Pull complete
d4dcde3aeeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
6376488be516: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-22-93 ~]$
```

ip-172-31-22-93.ec2.internal | 0% | 0.41 GB / 0.90 GB | 0.02 Mb/s | 0.00 Mb/s | 12 min | ec2-user (x2) | /: 27% | /tmp: 0% | /boot/efi: 13%

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1.4

Not secure 54.161.15.213

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

2.1

```
[ec2-user@ip-172-31-22-93 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-22-93 ~]$ newgrp docker
[ec2-user@ip-172-31-22-93 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                NAMES
a7e0a35e6584   nginx     "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes   0.0.0.0:80->80/tcp, :::80->80/tcp    great_edison
[ec2-user@ip-172-31-22-93 ~]$ docker volume create aribah
aribah
[ec2-user@ip-172-31-22-93 ~]$ docker volume ls
DRIVER    VOLUME NAME
local     aribah
[ec2-user@ip-172-31-22-93 ~]$
```

```
[ec2-user@ip-172-31-22-93 ~]$ docker run -it --aribah-container -v aribah:/data ubuntu bash
unknown flag: --aribah-container
See 'docker run --help'.
[ec2-user@ip-172-31-22-93 ~]$ docker run -it --name aribah-container -v aribah:/data ubuntu bash
root@439344be8f4d:/# ls /
bin  boot  data  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@439344be8f4d:/# echo "Name: Aribah Shirgaonker" > /data/info.txt
root@439344be8f4d:/# echo "Roll No. 24108A0016" >> /data/info.txt
root@439344be8f4d:/# cat /data/info.txt
Name: Aribah Shirgaonker
Roll No. 24108A0016
root@439344be8f4d:/#
```

3.1

```
[ec2-user@ip-172-31-22-93 ~]$ nano Dockerfile
[ec2-user@ip-172-31-22-93 ~]$ cat Dockerfile
FROM alpine

CMD ["echo", "Cloud Computing practical exam"]
[ec2-user@ip-172-31-22-93 ~]$
```

ip-172-31-22-93.ec2.internal | 0% | 0.40 GB / 0.90 GB | 0.02 Mb/s | 0.00 Mb/s | 38 min | ec2-user | /: 28% | /tmp: (

3.2

```
[ec2-user@ip-172-31-22-93 ~]$ docker build -t cloud-exam .
[+] Building 0.8s (5/5) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 156B
 => [internal] load metadata for docker.io/library/alpine:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/1] FROM docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4
 => => resolve docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4
 => => extracting sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84feafc269a76e3f96a63
 => => sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4 9.22kB / 9.22kB
 => => sha256:33154315cf4402e697f065e6ec2156e292187e633908ccfede9c66279b6fa956 1.02kB / 1.02kB
 => => sha256:772bee4540c411d1fec755cb8ead76db4173d38afde7b2018a08f255907bbf23 608B / 608B
 => => sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84feafc269a76e3f96a63 3.87MB / 3.87MB
 => exporting to image
 => => exporting layers
 => => writing image sha256:e5301451a74a00c30607e9e6a360094a520eb49e0689a42646be1ac2f1747c3f
 => => naming to docker.io/library/cloud-exam
[ec2-user@ip-172-31-22-93 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    936fef290c8f   2 days ago     161MB
cloud-exam    latest    e5301451a74a   3 days ago     8.45MB
ubuntu        latest    30ba44506a6d   7 weeks ago    100MB
hello-world   latest    e2ac70e7319a   2 months ago   10.1kB
[ec2-user@ip-172-31-22-93 ~]$
```

3.3

```
[ec2-user@ip-172-31-22-93 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    936fef290c8f   2 days ago     161MB
cloud-exam    latest    e5301451a74a   3 days ago     8.45MB
ubuntu        latest    30ba44506a6d   7 weeks ago    100MB
hello-world   latest    e2ac70e7319a   2 months ago   10.1kB
[ec2-user@ip-172-31-22-93 ~]$ docker run cloud-exam
Cloud Computing practical exam
[ec2-user@ip-172-31-22-93 ~]$
```

3.4	Difference between a Docker Image and a Docker Container.
	Docker image is the read only template also known as recipe, while container is the running instance of the image. If image is the recipe container is the finished dish.
3.5	Why are Docker Volumes used?
	Volumes are used to store data even if the container is deleted data stays in the volume. This keeps the data safe and also allows it to attach to other conatiners.